DA623: Computing with Signals

Mid-Semester Examination | Jan-May 2024, IIT Guwahati

·Total Questions: Eight (8) Total Points: Forty (40)
·Thought of the day - "Be calm and answer the questions". Time: 120 mins
·Write with clarity, stating any assumptions you make.

Question 1 [$1 \times 6 = 6$ points]

State True or False, and the reason for it.

(a) Classifier A is better than Classifier B if the training error of Classifier A is better than that of Classifier B.

(b) $y = \sum_{p=0}^{P} a_p x^p$ is an example of "generalized" linear regression for any finite P.

(c) $\|\mathbf{x}\|=x_1$ for $\mathbf{x} \in \mathbb{R}^3$ is a valid norm. Here, x_1 corresponds to the first element in $\mathbf{x}$.

(d) Consider $x(t) = \sum_{k=1}^{10} k \sin 2\pi k f_0 t$. The Nyquist sampling rate for this signal is $2f_0$.

(e) The design of DFT bases is independent of the data vector it is to be applied on.

(f) A straight forward implementation of DFT requires $N\mathcal{O}(N)$ operations.

Question 2 [$6 + 2 + 1 = 9$ points]

After landing on the Moon, Timmi needs to transmit ambient environment temperature data back to Earth. She opts to record temperature readings at hourly intervals. Following two lunar days of data collection, Timmi decides to do computational analysis to discern patterns.

(a) To begin, she seeks to devise a polynomial regression formulation using a pth−degree polynomial to fit the recorded data. State the formulation Timmi can use for this. This should include the regression equation, the cost function, and the equation for the final solution.

(b) Provide a procedure to choose p.

(c) On plotting the scatter plot of time versus temperature, Timmi observes the data points exhibit a periodic behavior. Comment on the periodicity (that is, time period) we should expect in this behavior.

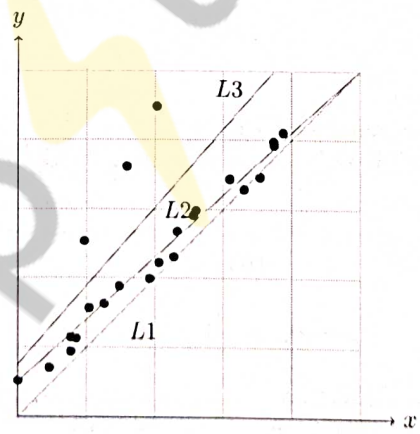

Fig. 1. The data points are distributed on the x-y plane. The x and y axis intersect at $(0, 0)$.

Question 3 [$2 \times 3 = 6$ points]

With regard to the regression problem depicted in Figure 1, label each of the following cost functions with the curve (that is, $L1, L2$, or $L3$) that minimizes it. State the reasons for each choice made.

(a) $\sum_{i=1}^{20} |y_i - x_i w_1 - w_0|$

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(b) $\sum_{i=1}^{20} |y_i - x_i w_1 - w_0|^2$

(c) $\sum_{i=1}^{20} |y_i - x_i w_1 - w_0|^2 + 10^5 w_0^2$

Question 4

[$2 \times 2 = 4$ points]

Robert has gathered data representing the trajectory traced by the seconds' hand of a traditional analog clock (that is, x and y coordinates). This clock's seconds' hand ticks once per second, adding a data sample at each second.

(a) After observing this trajectory for a complete 4-minute cycle, state (in terms of DFT bin and Hz) at which location on the spectrum axis will a peak occur for data corresponding to x-axis?

(b) State the above but for the data corresponding to the y-axis.

Question 5

[4 points]

Consider Brad has received a data vector which on plotting appears as shown in Figure 2 ($x[n]$ shown in the left panel). Brad has to devise a linear signal processing pipeline which when operated on $x[n]$ should output $y[n]$. Demonstrate your understanding of linear signal processing by helping Brad obtain $y[n]$ using one or more convolution operations. Specify the convolution signal(s) you will use.

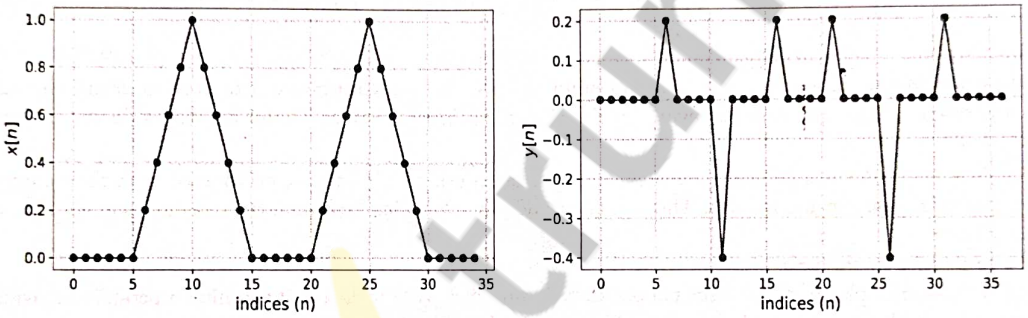

Fig. 2. Illustration of a data vector ($x[n]$, shown on left) and the processed data vector ($y[n]$, shown on right).

Question 6

[$2 \times 2 = 4$ points]

James has received $x(t)$ illustrated in Figure 3.

(a) Can you help James obtain the Fourier transform of $x(t)$.

(b) State with reasons if James sample $x(t)$ based on Shannon sampling theory and perfectly reconstruct it from the uniform samples.

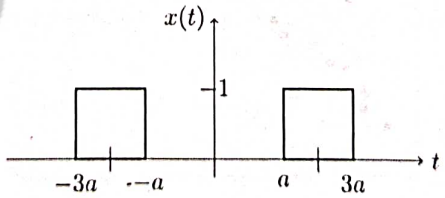

Fig. 3. Illustration of $x(t)$.

Question 7

[3 points]

You have three vectors: $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$. Provide a systematic approach to create an orthogonal basis using these vectors.

Question 8

[4 points]

Suppose you have a database containing a collection of palindromes. Devise (and state) an approach to learn the bi-gram probabilities from this database, and then utilize this information to synthesize new palindromes.

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DA623: Computing with Signals

End-Semester Examination | Jan-May 2024, IIT Guwahati

·Total Questions: ~~Eight~~ 4 (8)
·Thought of the day - "May the force be with you.".
·Write with clarity, stating any assumptions you make.

Total Points: Forty (40)
Time: 180 mins

Question 1

[~~4 × 2.5~~ 5×2 = 10 points]

Jimmy, a researcher in a biochemistry lab, has been provided with two datasets labeled A and B, which contain measurements of cell length and width obtained from experiments conducted under distinct conditions. Upon creating scatter plots of the data (as depicted in Fig 1), Jimmy seeks clarity on the following points. Assist Jimmy with answers.

(a) Determine which dataset exhibits a positive correlation between cell length and width?
(b) Identify which dataset demonstrates a higher variance in cell width?
(c) Determine which dataset showcases a lower mean value for cell length?
(d) Determine the most suitable 2-D probability distribution to model the data in each dataset.
(e) Comment on the feasibility of building a classifier to distinguish between cells belonging to Dataset A and Dataset B.

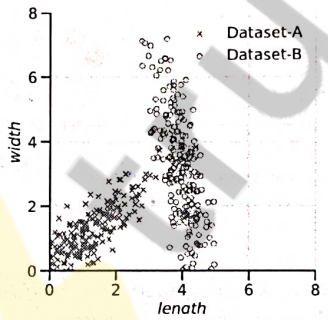

Fig. 1. Scatter plot for Question 1

Question 2

[5×2 = 10 points]

Wolfgang has received an audio signal containing different kinds of sound recordings stitched together. Being aware of the importance of short-time features in audio signal analysis, Wolfgang computes a spectrogram of the audio signal (see Fig 2). The temporal segments corresponding to the different sound categories contained in the audio signal are labelled as $T1$ to $T6$. Assist Wolfgang in mapping the below sound categories to the temporal segments $T1$ to $T6$. comment with reasoning for each mapping.

(a) fricative
(b) male speech
(c) female speech
(d) bird chirp
(e) DTMF tone

Assign one temporal segment to each of the sound category.

Question 3

[4 × 2.5 = 10 points]

Dave has received the dataset shown in Fig.3.

(a) What is the mean squared **leave one sample out** (during training) cross-validation error using linear regression? Assume the linear regression model as $y = \beta_0 + \beta_1 x$.
(b) What is the mean squared **leave one sample out** (during training) cross-validation error for a model $y = \beta_0$.
(c) What order polynomial fitting (based on least squares error minimization) to this dataset can result in over-fitting.
(d) Dave is progressively receiving additional data points. In this context, elaborate on how Dave can use $k-$fold cross-validation and also be sensitive to the aspect of bias-variance trade-off in choosing the polynomial order.

Leave the answers as fractions. Use of calculator is not desired.

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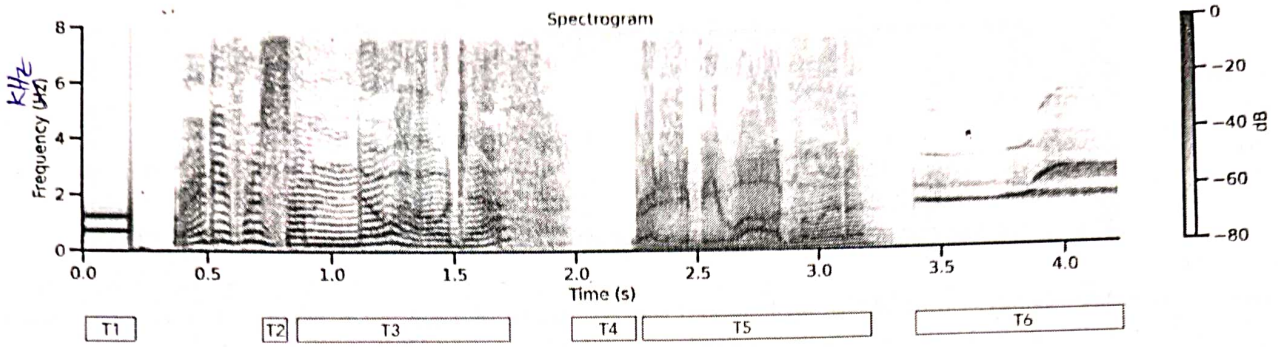

Fig. 2. Depiction of spectrogram of an audio signal. The audio signal had a sampling rate of 16 kHz in time-domain. The spectrogram was obtained by applying DFT to short-time segments of length 30 msec, and hopping with jump of 10 msec. The segments $T1$ to $T6$ denote time segments of interest in Question 2.

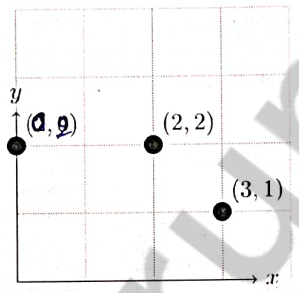

Fig. 3. The data points are distributed on the x-y plane. The x and y axis intersect at $(0, 0)$.

QUESTION 4

[$4 \times 2.5 = 10$ POINTS]

(a) Draw the image intensity histogram for an 8×8 chessboard pattern with alternating black and white squares image. Subsequently, apply a 2×2 smoothing kernel $[1\ 1; 1\ 1]/4$ using convolution (assuming mirroring of pixels at the borders) on the pixel intensities. Draw the resulting image intensity histogram. State the reason for dividing the smoothing kernel by a normalizing factor (4 in this case).

(b) Given a color image (with R,G, and B channels), comment on if the 2-D Fourier spectrum will be same or different for each channel.

(c) Devise a 3×3 kernel which can help in high-pass filtering an image. State the reasoning behind your design.

(d) Treating the spectrogram (see Fig.2) corresponding to the segment $T3$ as a 2-D matrix (or image), can you devise an approach to estimate the fundamental frequency corresponding to the pitch harmonics using 2-D Fourier transform.

All the best. Answer all the questions.

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